Learning Algebraic Invariants of Finite p-Groups from Commuting Graphs

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Abstract

We study how much information about a finite p-group survives in its commuting graph, and how well graph neural networks predict group invariants from it. For all 2668 nontrivial 2-groups of order at most 128 we build commuting graphs with node features (element order, centralizer and conjugacy class sizes, membership of the centre and power depth) and train Graph Isomorphism Networks to predict nilpotency class, derived length, centre size, commutator subgroup size and the number of conjugacy classes. Two of these targets turn out to be closed-form functions of the node features. An exhaustive census of collisions, that is, non-isomorphic groups with isomorphic commuting graphs, shows that the 2668 groups have only 75 distinct commuting graphs. It gives exact lower bounds on the error of any model that reads the graph alone (R_0), with element orders (R_1), or with element orders and power depth (R_2). Building on the known fact (Pezzott and Nakaoka) that isoclinic groups of the same order have isomorphic commuting graphs, we show that the exponent and powerfulness are not determined by the commuting graph, even for non-abelian groups, and that only non-isoclinic collisions can separate any of the five targets. Such collisions occur for commutator size and for nilpotency class, the latter already at order 32, where $C_2 \times D_{16}$ and $(C_2 \times C_2) \wr C_2$ share a commuting graph. No collision in our datasets separates derived length. Gradient boosting on per-graph summary statistics outperforms the GNN on every target, and, averaged over five seeds, the GNN is worse than predicting the training mean on two targets, mainly because of one failed seed. We also extend the study to powerful and powerfully nilpotent p-groups, following Traustason and Williams, and to the 594 nontrivial 3-groups of order at most 729.

1 Introduction

The classification and analysis of finite p-groups is a central topic in group theory. Although complete classifications are available for small orders through databases such as the SmallGroups library [2], the number of isomorphism types grows very fast. There are 2328 groups of order $128 = 2^7$, 56092 of order $256 = 2^8$, and already more than ten million (10 494 213) of order $512 = 2^9$. The SmallGroups library does not contain the groups of order 1024 [2].

Computational tools such as GAP [1] compute the invariants studied here directly from the group, and the commuting graph is itself built from the group, so learning them from the graph is not a computational shortcut. The question we study is how much information about a finite p-group survives in a graph derived from its multiplication structure, in particular its commuting graph: which invariants the graph determines, which it does not, and how well a model that reads the graph predicts them.

The vertices of such a graph carry no canonical order, so we use graph neural networks (GNNs), which are invariant under relabelling of the vertices. Our network has GIN convolutional layers [3] and a mean-pooled readout. The maximal Weisfeiler–Lehman expressiveness guarantee of [3] requires a sum-based readout, so it does not apply to our model (Section 4).

Weissblat trains GNNs on Cayley graphs of finite groups to predict solvability [27], and extends this to a common Cayley-graph framework covering abelianity, nilpotency, and solvability [28]. Barket et al. compare classical models, MLPs, and GNNs across a census of more than 131 000 Cayley graphs of groups of order up to 767, and find engineered graph statistics highly informative [29]. The baseline results of Section 7.2 are consistent with this. Earlier, He and Kim investigated learning algebraic structures directly from multiplication tables [30]. Eick and Schanze study, by algorithmic methods and without learning, which invariants suffice to distinguish among the 10 494 213 groups of order 512 [31]. Their question has the same flavour as the recoverability framework of Section 5. The present work differs in using commuting graphs rather than Cayley graphs, and in computing exact collision classes and representation-specific error floors before any model is trained.

The commuting relation encodes centralizers, abelian subgroups, and conjugacy information, and we enrich the commuting graph with additional features adapted to the power structure of 2-groups.

Classical results show that commuting graphs do not uniquely determine the group (Section 2.3). Section 5 is organized around the resulting question of how much information is lost in passing to the graph, and which invariants can still be recovered from it.

A further motivation comes from the theory of powerful and powerfully nilpotent p-groups [12, 13, 14, 15, 16, 17], where characteristic filtrations (the power series and the upper powerfully central series) are standard tools. These filtrations suggest node features (the power-depth feature below) and a new prediction target, the *powerful nilpotence class* of Traustason and Williams [14], discussed in Section 8.

1.1 Contributions

Our contributions are:

- An exhaustive census of commuting-graph collisions among all nontrivial 2-groups of order at most 128 and all nontrivial 3-groups of order at most 729, at three representation levels (R_0, the plain commuting graph; R_1, with element-order labels; R_2, with power-depth labels as well), together with the resulting exact Bayes floors per target (Proposition 5.17). The 2668 2-groups have only 75 distinct commuting graphs. Collisions separate nilpotency class from order 32 onwards, and we give an explicit pair at that order. Some of them, at order 128, survive even the R_2 labels. No collision in this range separates derived length.

- A dataset of enriched commuting-graph representations of all nontrivial 2-groups of order at most 128, together with the classical invariants of Section 2 and, optionally, whether each group is powerful and whether it is powerfully nilpotent, and if it is, its powerful nilpotence class and powerful coclass in the sense of [14]. The construction takes the prime p as a parameter, so it also applies at odd primes.

- A GIN-based model that combines algebraic node features with an optional persistent-homology branch on the noncentral commuting graph. In the single-seed comparison we ran, adding the branch did not improve the errors (Section 7.2).

- A formal framework for collisions and recoverability (Section 5) in which we can state which invariants are determined by a given graph representation, and closed-form baselines (Propositions 5.7 and 5.8) for two of our five targets, which we use to separate structural learning from feature lookup in the results.

- A collision analysis for commuting graphs built on known results. Same-order isoclinic groups have isomorphic commuting graphs (Pezzott and Nakaoka [20]; Theorem 5.11, proof included for completeness), and the two non-abelian groups of order p^3 share the graph $K_p \vee ((p+1) \cdot K_{p^2-p})$ [21, 24] (Proposition 5.12). We derive the consequences for our setting: neither the exponent nor powerfulness is recoverable from the plain commuting graph (Corollary 5.13), $\tau(\exp) = 1$ (Corollary 5.14), and any collision separating one of the five classical targets must be non-isoclinic (Corollary 5.15). The known order-128 pair of Arvind and Cameron [25] is such a collision, and it separates commutator size (Corollary 5.16).

- An extension to powerful and powerfully nilpotent p-groups, implemented and evaluated empirically (Section 8). It includes a worked example (Proposition 8.1), which we use as a check on our computations, of a group that is powerful but not powerfully nilpotent at an odd prime (a known phenomenon [17, Sec. 4.1]); a structural link (Theorem 9.1, Corollary 9.2) between our power-depth node feature and Williams' work on normal subgroups of powerful p-groups [15]; and a proposal to use the p-spectrum of [18] as a further, structurally different prediction target (Section 8.3).

- An order-transfer experiment at $p = 2$ (train on orders ≤ 64, evaluate zero-shot on order 128) and a single-seed cross-prime fine-tuning experiment from $p = 2$ to $p = 3$, both on the full datasets (Section 10).

2 Background

2.1 Finite p-Groups

Let p be a prime. A finite group G is a p-group if $|G| = p^n$ for some n.

Every finite p-group is nilpotent and admits a lower central series

$$\gamma_1(G) = G, \qquad \gamma_{i+1}(G) = [\gamma_i(G), G].$$

The smallest c such that $\gamma_{c+1}(G) = \{e\}$ is the nilpotency class (nontrivial abelian groups have class 1, and the trivial group has class 0).

Further fundamental invariants include the derived length, $|Z(G)|$, $|[G, G]|$, the number of conjugacy classes $k(G)$, and the exponent $\exp(G)$.

2.2 Powerful and Powerfully Nilpotent p-Groups

A finite p-group G is *powerful* if $[G, G] \leq G^p$ for p odd, or $[G, G] \leq G^4$ for $p = 2$, where $H^{p^k} := \langle x^{p^k} : x \in H \rangle$ is generated by the p^k-th powers of all elements of H [12, 13]. The p^k-th powers of a generating set of H need not generate the same subgroup. Powerful p-groups behave in many ways like abelian groups. Every powerful 2-group of order at most 8 is abelian, and non-abelian powerful 2-groups first appear at order 16 (e.g. the modular group of order 16).

Traustason and Williams [14] introduce *powerfully nilpotent* groups: powerful p-groups possessing an ascending chain $\{1\} = H_0 \leq H_1 \leq \cdots \leq H_n = G$ that is *powerfully central*, meaning $[H_i, G] \leq H_{i-1}^p$ for each i. The shortest such chain length is the *powerful nilpotence class* of G, and if $|G| = p^m$ and G has powerful class c, then $m - c$ is the *powerful coclass*. The fastest such chain is the *upper powerfully central series*

$$\hat{Z}_0(G) = \{1\}, \qquad \hat{Z}_n(G) = \{a \in G : [a, x] \in \hat{Z}_{n-1}(G)^p \text{ for all } x \in G\},$$

and the powerful class is the smallest n with $\hat{Z}_n(G) = G$. If this series terminates strictly below G, then G, although powerful, is not powerfully nilpotent. The series can be computed in GAP from this definition, and we compute the powerful class this way (Section 8).

Every powerful 2-group is powerfully nilpotent [14], a fact Williams uses in treating the case $p = 2$ [15]. This is special to $p = 2$. In [14, 17] the powerfully nilpotent groups are introduced as "a special subclass" of the powerful p-groups, and in general they form a proper subclass. Section 8 below gives an odd-prime example where the two notions differ.

A p-group is *strongly powerful* if $[G, G] \leq G^{p^2}$. Traustason and Williams show a strongly powerful group is automatically powerfully nilpotent, of powerful class at most $e-1$ where p^e is the exponent and $e \geq 2$ [14] (see also [17]). For 2-generator groups the converse also holds:

Proposition 2.1 ([17, Lemma 2.1])**.** *Every 2-generator powerfully nilpotent p-group is strongly powerful.*

This gives a necessary condition that is easy to check, and we use it in Section 8. A 2-generator powerful group that is not strongly powerful cannot be powerfully nilpotent, independent of any computation of $\hat{Z}_n(G)$.

Traustason and Williams [17] give a full classification of the non-abelian 2-generator ("rank 2") powerfully nilpotent groups (every abelian p-group is powerfully nilpotent), in two families determined by parameters n, m, r (or n, m, l, r):

$$G(n, m, r) = \langle a, b \mid a^{p^n} = b^{p^m} = 1,\ [a, b] = a^{p^r} \rangle, \qquad n - r \leq m,\ 2 \leq r \leq n - 1,$$

("split", i.e. a semidirect product of $\langle a \rangle$ by $\langle b \rangle$) together with a "non-split" family $G(n, m, l, r)$, in which the relation $b^{p^m} = 1$ is replaced by $b^{p^m} = a^{p^l}$, with parameter constraints $2 \leq r < l \leq n - 1$, $n - r \leq l < m$. Their classification of all powerfully nilpotent p-groups of order up to p^6, carried out in [17, Sec. 4] case by case for p odd and $p = 2$ separately, also uses two families $A(n, t, s)$ and $B(n, t, s)$ of groups of type $(1^t, n)$, generated by t elements of order p and one element of order p^n, together with a short list of exceptional groups. They organize the powerfully nilpotent groups into an *ancestry tree*, in which a non-abelian powerfully nilpotent group G is a direct ancestor of $G/Z(G)^p$.

Frequency of powerfully nilpotent 2-groups. For $p = 2$, [17, Sec. 4.2] lists every non-abelian powerfully nilpotent 2-group of order 16, 32, and 64 (one group of order 16, four of order 32, and twelve of order 64). We compare these counts with the total number of 2-groups at each order in Table 1.

Order	Total 2-groups	Powerfully nilpotent	Fraction
8	5	3 (all abelian)	60%
16	14	6 (5 abelian + 1 non-abelian)	42.9%
32	51	11 (7 abelian + 4 non-abelian)	21.6%
64	267	23 (11 abelian + 12 non-abelian)	8.6%

Table 1: Powerfully nilpotent 2-groups as a fraction of all 2-groups of that order, from the classification in [17, Sec. 4.2]. The non-abelian groups D_8 and Q_8 of order 8 are not powerful. At $p = 2$ every powerful group is powerfully nilpotent, and the numbers of powerful groups in our dataset (Section 8) agree with every row of the table. At order 128 the fraction falls further, to $51/2328 \approx 2.2\%$.

For the rank-2 family alone, [17, Prop. 2.2] also gives a closed-form enumeration formula, which is cubic in the order exponent x on each residue class of x modulo 6.

2.3 Commuting Graphs

Let G be a finite group. The commuting graph $\Gamma(G)$ has vertex set G, with an edge between distinct elements x, y if and only if $xy = yx$. Elements of the centre commute with everything and so correspond to universal vertices in $\Gamma(G)$. Some treatments of commuting graphs exclude the centre from the vertex set, since a universal vertex says little about the structure of the graph. We include the centre, and membership of the centre is also recorded as a node feature. One consequence is that the graph is complete for every abelian group, and Lemma 3.3 (Section 3) shows that its persistent homology is trivial for every group.

Non-isomorphic groups may have isomorphic commuting graphs. All abelian groups of the same order have the same (complete) commuting graph, and so do D_8 and Q_8 [23]. Isoclinism gives a systematic source of such collisions (Section 5.5). This is a special case of the general phenomenon that group-to-graph encodings can be many-to-one (see Section 5).

$\Gamma(G)$ records only the binary relation "$[x, y] = e$". It does not record finer information such as the order of $[x, y]$ when x and y do not commute. Recording this would require additional edges or edge weights, or a different representation (Section 10).

2.4 Isoclinism

Two finite groups G and H are *isoclinic* (Hall [11]) if there are isomorphisms $\alpha : G/Z(G) \to H/Z(H)$ and $\beta : [G, G] \to [H, H]$ that are compatible with the commutator maps: whenever $\alpha(xZ(G)) = x'Z(H)$ and $\alpha(yZ(G)) = y'Z(H)$, we have $\beta([x, y]) = [x', y']$. Isoclinism is the equivalence Hall introduced to organize the classification of p-groups. We use the following classical facts. Nontrivial isoclinic groups have equal nilpotency class and derived length, and isomorphic derived subgroups. All abelian groups are isoclinic to the trivial group, and hence to one another. The groups D_8 and Q_8 are isoclinic, as are the two non-abelian groups of order p^3 for every prime p.

2.5 Graph Neural Networks

Graph neural networks iteratively aggregate neighbourhood information to produce node and graph embeddings. Such message-passing networks cannot distinguish more graphs than the 1-dimensional Weisfeiler–Lehman test (1-WL) [3, 4]. We use GIN convolutional layers [3] followed by global mean pooling. The maximal Weisfeiler–Lehman expressiveness guarantee of [3] requires injective, sum-based aggregation and readout. Our mean-pooled readout does not have this guarantee. Sum pooling and the concatenation of mean and sum pooling are also implemented.

3 Dataset Construction

3.1 Group Generation

We enumerate all groups of order 2^n for $1 \le n \le 7$ using GAP's SmallGroups library [2], via the script `generate_groups_sage.py`. This gives $1 + 2 + 5 + 14 + 51 + 267 + 2328 = 2668$ groups in total. For each group G we compute the classical invariants of Section 2, plus, optionally, whether G is powerful and, if so, its powerful nilpotence class and powerful coclass (Section 8).

Remark 3.1. The split into training, validation, and test sets is stratified by order. Orders with fewer than 10 groups are placed entirely in the training set, because a 10% slice of 1, 2, or 5 groups cannot populate every split. The resulting split sizes for each order can be listed with `report_split_by_order()` in `dataset.py`.

3.2 Graph Representation

Each group is represented by a graph whose vertices correspond to the elements of G, with edges given by the commuting relation (Section 2.3).

Node features. Each vertex $x \in G$ is assigned a 5-dimensional feature vector

$$(|x|,\ |C_G(x)|,\ |\mathrm{Cl}(x)|,\ \mathbf{1}_{x \in Z(G)},\ \mathrm{pdepth}(x)),$$

where $\mathrm{pdepth}(x)$ is the largest $k \geq 0$ such that $x \in G^{2^k} = \langle g^{2^k} : g \in G \rangle$, with g running over all elements of G. On an irregular 2-group the subgroup generated by the 2^k-th powers of a generating set need not equal G^{2^k}. The identity lies in every power subgroup. By convention its depth is the terminal filtration index e with $G^{2^e} = \{1\}$, i.e. $e = \log_2 \exp(G)$.

Edges. Edges are unweighted. A possible extension would add the commutator-magnitude edge attribute $w(x, y) = \mathrm{ord}([x, y])$ and use an edge-aware GIN variant. Edges join only commuting pairs, where $[x, y] = e$, so this attribute would be constant and uninformative unless the graph is enlarged to include non-commuting pairs. This extension was not evaluated in this study and is discussed in Section 10.

3.3 Persistent Homology

We optionally compute persistent homology of the commuting graph's shortest-path metric using Vietoris–Rips complexes, via giotto-tda [10]. Persistence diagrams in dimension 1 are converted to 10×10 persistence images, giving a 100-dimensional graph-level feature vector.

Remark 3.2. The Vietoris–Rips complexes are built from the matrix of shortest-path distances, which is passed to giotto-tda as a precomputed metric (`metric="precomputed"`), so its rows are not read as the coordinates of a point cloud. For disconnected noncentral graphs, infinite inter-component distances are replaced by one plus the largest finite distance before the transform is applied.

The next lemma shows that persistent homology of the full commuting graph gives no information about G beyond its order.

Lemma 3.3 (PH-triviality of the full commuting graph)**.** *Let G be a nontrivial finite group and equip $\Gamma(G)$ with its shortest-path metric. For every threshold t, the Vietoris–Rips complex $\mathrm{VR}_t(\Gamma(G))$ is a disjoint union of points ($t < 1$) or contractible ($t \geq 1$). Hence $H_k(\mathrm{VR}_t) = 0$ for all $k \geq 1$ and all t, and every persistent-homology descriptor of the full commuting graph is the same for every group of a given order.*

Proof. The identity commutes with everything, so it is a universal vertex and all pairwise distances are 1 or 2. For $1 \leq t < 2$, VR_t is the clique complex of $\Gamma(G)$. Every clique remains a clique after adjoining the identity, so the complex is a cone with apex the identity, hence contractible. For $t \geq 2$ the complex is a full simplex. □

We therefore compute persistent homology on the *noncentral* commuting graph (all vertices of $Z(G)$ removed, using the centrality node feature). This is a simple modification under which contractibility is no longer automatic. For example, the noncentral graph of $D_8 \times D_8$ contains induced 4-cycles. Whether H_1 is nontrivial for a given group has to be checked by computation, and we report the fraction of groups with nontrivial noncentral H_1 as a diagnostic (computed in `dataset.py`). For abelian groups the noncentral graph is empty and the descriptor is zero by convention. One `PersistenceImage` transformer is fitted on the training diagrams only and applied to all splits, so image grids are consistent across groups and no test information leaks into the transform.

4 Neural Network Architecture

Our model consists of three GIN message-passing layers (hidden dimension 128), global mean pooling, concatenation with the 100-dimensional PH feature vector when enabled (228-dimensional pooled representation), and two fully connected layers producing a 5-dimensional output, one per regression target (`PGroupGNN` in `model.py`).

5 A Framework for Recoverability of Group Invariants from Graph Representations

We ask how much algebraic information about a finite p-group can be recovered from graph-based representations derived from its internal structure.

> *Given a representation of a group G by a graph or enriched graph structure $R(G)$, which algebraic invariants of G are determined by $R(G)$, and which require additional information?*

5.1 Graph Representations of Finite p-Groups

Let $\mathcal{C}$ denote a class of finite p-groups. A *graph representation* of groups in $\mathcal{C}$ is a mapping $R : \mathcal{C} \to \mathcal{G}$, where $\mathcal{G}$ denotes a class of graphs (possibly with node or edge labels). Examples include the commuting graph, the noncentral commuting graph, and graphs enriched with node or edge labels as in Section 3.2.

5.2 Exact Recoverability of Invariants

Let $I : \mathcal{C} \to \mathcal{X}$ be a group invariant.

Definition 5.1 (Exact Recoverability)**.** An invariant I is *exactly recoverable* from a representation R on $\mathcal{C}$ if for all $G, H \in \mathcal{C}$, $R(G) \cong R(H) \Rightarrow I(G) = I(H)$.

Definition 5.2 (Collision)**.** A pair $G, H \in \mathcal{C}$ is an *R-collision for I* if $R(G) \cong R(H)$ but $I(G) \neq I(H)$.

Definition 5.3 (Recoverability Threshold)**.** For a hierarchy of representations $R_0, R_1, R_2, \ldots$ each containing at least as much information as the last, $\tau(I) = \min\{k : I \text{ is recoverable from } R_k\}$.

Throughout, we use the following hierarchy

$$R_0 = \Gamma(G) \text{ (unlabelled)}, \quad R_1 = R_0 + \text{element-order node labels}, \quad R_2 = R_1 + \text{pdepth node labels}.$$

The remaining node features of Section 3.2 are centralizer size, class size and centrality. All three are R_0-determined (Propositions 5.5–5.7), so the full feature set used in our experiments lies at level R_2. The persistent-homology features are functions of $\Gamma(G)$ alone. They can affect statistical learnability but cannot change exact recoverability at any level of the hierarchy, so questions of exact recoverability concern only the node labels.

5.3 Invariants Determined by the Commuting Graph

The facts in this subsection are classical. The centralizer–degree correspondence goes back to the Brauer–Fowler tradition (see [24] and the survey [23]). We include the short arguments, since Section 5.5 relies on them.

Proposition 5.4. *$|G|$ is determined by $\Gamma(G)$: it is the number of vertices.*

Proposition 5.5. *$|C_G(x)|$ is determined by $\Gamma(G)$: $|C_G(x)| = \deg(x) + 1$.*

Proposition 5.6. *$|x^G|$ is determined by $\Gamma(G)$: by orbit–stabilizer, $|x^G| = |G|/|C_G(x)|$, and both quantities on the right are determined by the graph.*

Proposition 5.7 (Centre size)**.** *$|Z(G)|$ is determined by $\Gamma(G)$: an element lies in the centre iff it is adjacent to every other vertex.*

Proposition 5.8 (Number of conjugacy classes)**.** *$k(G)$ is determined by $\Gamma(G)$: $k(G) = \frac{1}{|G|}\#\{(x, y) \in G^2 : xy = yx\}$, equivalently (Burnside's lemma applied to the conjugation action) $k(G)$ is the average centralizer size, $k(G) = \frac{1}{|G|}\sum_{x\in G} |C_G(x)|$.*

Remark 5.9 (Centre size and $k(G)$ in Table 2). By Propositions 5.7 and 5.8, centre size and $k(G)$ can be read off from two of our node features, $\mathbf{1}_{x\in Z(G)}$ and $|C_G(x)|$. We have

$$|Z(G)| = \sum_x \mathbf{1}_{x\in Z(G)}, \qquad k(G) = \text{mean}_x\, |C_G(x)|,$$

so both targets are exact functionals of supplied input features. Our readout is global mean pooling, and $k(G)$ equals this readout applied to the centralizer-size column, so no message passing is needed. Centre size is a sum, which mean pooling recovers only after rescaling by $|G|$. The order $|G|$ is itself recoverable from the representation (it is the centralizer-size feature of any central vertex), but not by a single trivial readout. The closed-form baseline (computed in `train.ipynb`) accordingly uses add-pooling for centre size and mean-pooling for $k(G)$. Low error on these two targets is not evidence of deeper structural learning. We verified both identities computationally on thirteen hand-constructed test groups of orders 4 to 27, and they held to floating-point precision. Table 2 therefore reports these two closed-form values in its second column. On these two targets the GNN should be judged against that column.

5.4 Invariants Not Obviously Determined

Nilpotency class, derived length and coclass are not obviously determined by $\Gamma(G)$ alone. The minimal number of generators and the order of the Frattini subgroup are not determined, since Γ is complete for every abelian group and C_4 and $C_2 \times C_2$ differ in both. We prove that the exponent and the property of being powerful are not determined by $\Gamma(G)$ even for non-abelian groups (Corollary 5.13 below).

5.5 Collisions via Isoclinism

This subsection collects known methods for constructing collisions. Commuting is invariant under central translation, so the entire commuting graph factors through $G/Z(G)$, which is one of the two ingredients that Hall's isoclinism (Section 2.4) preserves. Lemma 5.10, Theorem 5.11 and Proposition 5.12 below are known. Pezzott and Nakaoka [20, Prop. 2.3] prove that isoclinic non-abelian groups have isomorphic commuting graphs on a transversal of the centre, and describe how the commuting graph is recovered from this graph by replacing each vertex with its coset [20, Sec. 2]. Together these give Theorem 5.11, which Cameron and Kuzma [19, Sec. 2] and Arvind et al. [24, Thm. 9.1] state in this form, crediting Pezzott and Nakaoka. We include short proofs to keep the collision analysis self-contained. The corollaries apply these results to our five targets and to the representation levels R_0, R_1 and R_2. The new material in this subsection is the census below, which determines which collisions occur in our datasets and which targets they separate.

Lemma 5.10 (Blow-up structure)**.** *Let G be a finite group with centre Z. The relation $xZ \sim yZ \iff [x,y]=1$ is well-defined on G/Z. Write $\bar{\Gamma}(G)$ for the graph on G/Z with this adjacency. Then $\Gamma(G)$ is isomorphic to the lexicographic blow-up of $\bar{\Gamma}(G)$ in which every coset is replaced by a clique of size $|Z|$, two cliques being joined completely if and only if the corresponding cosets are adjacent.*

Proof. If $z, w \in Z$ then $[xz, yw] = [x, y]$, so the relation is well-defined, and two elements of the same coset always commute (they differ by a central element), giving the within-coset cliques. Adjacency between elements of distinct cosets depends only on the cosets, giving the complete joins. □

Theorem 5.11 (Isoclinic collisions)**.** *If G and H are isoclinic finite groups with $|G| = |H|$, then $\Gamma(G) \cong \Gamma(H)$.*

Proof. Isoclinism supplies $\alpha : G/Z(G) \to H/Z(H)$ and $\beta : [G,G] \to [H,H]$ with $\beta([x,y]) = [x',y']$ for representatives x', y' of $\alpha(xZ), \alpha(yZ)$. Since β is an isomorphism, $[x,y]=1 \iff [x',y']=1$, so α is an isomorphism $\bar{\Gamma}(G) \to \bar{\Gamma}(H)$. From $|G| = |H|$ and $|G/Z(G)| = |H/Z(H)|$ we get $|Z(G)| = |Z(H)|$. Blowing both reduced graphs up by the same clique size and applying Lemma 5.10 gives $\Gamma(G) \cong \Gamma(H)$. □

The theorem converts every same-order isoclinic pair into a collision. The simplest instances are abelian. All abelian groups are isoclinic, and Γ is complete for every abelian group, so e.g. C_4 and $C_2 \times C_2$ collide while having exponents $4 \neq 2$. The following family shows the phenomenon is not an artefact of complete graphs.

Proposition 5.12 (The order-p^3 family)**.** *Let p be a prime and G a non-abelian group of order p^3. Then*

$$\Gamma(G) \;\cong\; K_p \vee \big((p+1)\cdot K_{p^2-p}\big),$$

the join of a p-clique with $p{+}1$ disjoint cliques of size p^2-p. In particular the two non-abelian groups of order p^3 have isomorphic commuting graphs for every prime p.

The noncentral part of the graph, $(p+1)\cdot K_{p^2-p}$, appears in the proof of Theorem 2.1 of Dutta and Nath [21], who apply it to the non-abelian groups of order p^3 in their Corollary 2.2. For odd p, that the two groups have isomorphic commuting graphs is a case of [24, Lemma 7.2(a)], and for $p=2$ (the groups D_8 and Q_8) it is noted in [23].

Proof. For non-abelian G of order p^3 we have $|Z(G)| = p$ and $[G, G] = Z(G)$ (standard). For noncentral x, every conjugate $x^g = x[x, g]$ lies in xZ, so the class x^G has size at most p. It is not 1, and class sizes divide $|G|$, so $|x^G| = p$ and $|C_G(x)| = p^2$. A group of order p^2 is abelian, and $C_G(x)$ contains Z and x, so $C_G(x)$ is the preimage of the order-p subgroup $\langle xZ\rangle \leq G/Z \cong C_p \times C_p$. That quotient has exactly $p + 1$ subgroups of order p, so the $p^3 - p$ noncentral elements are partitioned into $p + 1$ abelian "lines" of $p^2 - p$ elements each. Elements within a line commute (the line is abelian), and noncentral elements of different lines do not (if they did, $C_G(x)$ would contain the strictly larger subgroup $\langle x, y, Z\rangle$). The central elements are universal vertices. This is the stated join. □

Corollary 5.13. *Neither the exponent nor the property of being powerful is exactly recoverable from $\Gamma(G)$, even on the class of non-abelian p-groups.*

Proof. Take $p = 3$ in Proposition 5.12: the Heisenberg group (exponent 3, not powerful since $G^3 = 1$ while $[G, G]$ has order 3) and $\mathbb{Z}_9 \rtimes \mathbb{Z}_3$ (exponent 9, powerful by Proposition 8.1) have isomorphic commuting graphs. □

We verified Proposition 5.12 computationally for $p = 2$ and $p = 3$, and Corollary 5.13 for $p = 3$: both order-27 groups produce the graph $K_3 \vee (4 \cdot K_6)$, and both order-8 non-abelian groups (D_8, Q_8) produce $K_2 \vee (3 \cdot K_2)$, with all five classical targets agreeing across each pair while the element-order multisets differ (`tests/test_group_theory.py`). Since a graph of the form "universal clique joined to disjoint cliques" is determined up to isomorphism by its clique sizes, these structure checks prove that the graphs are isomorphic.

Corollary 5.14 (Threshold for the exponent)**.** *In the hierarchy $R_0 \subset R_1 \subset R_2$, $\tau(\exp) = 1$: it is not 0 by Corollary 5.13, and it is at most 1 because the exponent of a p-group is the maximum element order, a function of the R_1 node labels. Similarly $\tau($is powerful$) \geq 1$.*

The census below shows that powerfulness is not determined at any level of this hierarchy: some groups in our datasets have commuting graphs that are isomorphic even with element-order and power-depth labels, yet differ in powerfulness.

Corollary 5.15 (Damaging collisions are non-isoclinic)**.** *Γ determines $|G|$, so every collision is between groups of equal order. Nilpotency class, derived length, and $|[G, G]|$ are isoclinism invariants (Section 2.4), and $|Z(G)|$ and $k(G)$ are Γ-determined (Propositions 5.7, 5.8). Hence if a collision pair is isoclinic, all five regression targets of this paper agree across it. Any commuting-graph collision that separates one of the five targets must therefore be a pair of non-isoclinic groups with isomorphic commuting graphs.*

Non-isoclinic collisions are known to exist, and one lies within the range of orders in our dataset. Arvind and Cameron [25] (see also [24, Sec. 9]), building on Example 3 of Cameron and Kuzma [19], take $G = \mathrm{SmallGroup}(64, 182)$ (nilpotency class 3). Its Schur covers H have order 128. They are the groups $\mathrm{SmallGroup}(128, i)$ with $i \in \{789, 790, 791, 815, 816, 817\}$, and they are commutation-preserving because the Bogomolov multiplier of G equals its Schur multiplier. It follows that $\Gamma(H)$ and $\Gamma(G \times C_2)$ are both the 2-clique blow-up of $\Gamma(G)$, and hence are isomorphic. However, H and $G \times C_2$ are not isoclinic. This gives a further non-recoverability result for one of our five targets.

Corollary 5.16 (Commutator size is not recoverable)**.** *On 2-groups of order 128, the commutator subgroup size is not exactly recoverable from $\Gamma(G)$. Indeed, for the pair above, the covering map $H \to G$ has central kernel $K \leq H'$ of order 2 and maps H' onto G', so $|H'| = 2\,|G'|$, while $|(G \times C_2)'| = |G'|$. The two groups collide but their commutator sizes differ by a factor of 2.*

By Corollary 5.15 this pair is necessarily non-isoclinic, which is how it is exhibited in [24]. By Proposition 5.17 it forces a strictly positive MSE floor for commutator size on every model whose input is the unlabelled graph R_0, independent of architecture and training. This argument does not settle whether the pair remains a collision under the R_1 or R_2 node labels, and hence whether the floor survives for the feature-enriched input of the trained network. That question is decided by the labelled refinement in the census (below). Arvind and Cameron [25] conjectured that class-2 nilpotent groups are determined up to isoclinism by their commuting graphs. The later version [24, Sec. 9] reports that Eamonn O'Brien found counterexamples of order 64, with catalogue numbers $(64, 18)$ and $(64, 215)$, refuting the conjecture. Non-isoclinic collisions therefore also occur in class 2, although this pair agrees on all five of our targets. The converse direction does hold for abelian groups, non-abelian simple groups (via Solomon–Woldar [26]), and extraspecial groups.

We examined the collisions in our dataset with the collision census (`src/collision_census.py`). The census first sorts the commuting graphs into buckets by order and Weisfeiler–Lehman hash [7] (equal hashes are a necessary condition for isomorphism). It then groups each bucket by BLISS canonical form [8, 9], which two graphs share if and only if they are isomorphic, and refines each confirmed class by labelled isomorphism, using label-seeded WL prebucketing confirmed by a pairwise coloured BLISS search. The labels are element orders for R_1, and element order together with power depth for R_2. Finally, the census computes the per-target floor of Proposition 5.17 at each of R_0, R_1 and R_2. The trained network receives R_2-level input, so only the R_2 floors bound its error. On the thirteen hand-constructed test groups, the census gives an exponent floor of 4.31 at R_0 and 0 at R_1, so the collisions responsible for that floor do not survive the element-order labels. Among these groups the census finds six confirmed collision classes, including the pair D_{16} and SD_{16}, a maximal-class collision at an order inside the range of the main dataset. None of the six separates any of the five targets, and in each of them the R_1 labels distinguish all the groups. Apart from the exponent's floor, the only positive R_0 floor is a $1/13$ classification floor for powerfulness. Both floors agree with what the theory predicts at these orders. The census also reports whether the O'Brien pair of order 64 appears among its collisions. As a check on the known pair of order 128, whenever the dataset contains groups of that order the census tests whether some collision class contains one of the known Schur-cover catalogue numbers, and warns if none does.

On the full dataset of 2668 groups, the census confirms 74 R_0 collision classes covering 2667 of them. The only group outside these classes is the unique group of order 2, so every other group shares its commuting graph with another group in the dataset. After refinement, 458 classes survive the R_1 labels and 360 survive the R_2 labels. The census took 36 s of wall-clock time using canonical-form grouping. Both literature validation cases are recovered: the Arvind–Cameron order-128 collision appears with its factor-2 commutator signature, and the reported O'Brien order-64 pair $(64, 18)/(64, 215)$ is confirmed as a collision. The census answers the questions left open above as follows.

- Nilpotency class: several non-isoclinic collision classes split it, giving Bayes floors of 0.1408 at R_0, 0.0090 at R_1 and 0.0005 at R_2. The floor is still positive at R_2, the representation that the trained network takes as input.

- Derived length: no collision class in this dataset separates it. Its floor is 0 at all three levels, so no obstruction to recoverability appears up to order 128 (this does not preclude collisions at larger orders).

- Commutator size: 0.0217 at R_0, 0.0066 at R_1 and 0.0047 at R_2. This confirms the proved positive R_0 floor numerically. The floor stays positive under enrichment, whereas the exponent's floor vanishes at R_1.

- Exponent: 17.60 at R_0, dropping to 0 at R_1, as Corollaries 5.13 and 5.14 predict.
- Powerfulness: 0.0199 at R_0, 0.0090 at R_1 and 0.0045 at R_2 (a 0–1 classification floor). The floors for conditional powerful class are 0.0103, 0 and 0.

The smallest collision class in the 2-group dataset that separates nilpotency class has order 32. It contains 19 of the 51 groups of that order, ten of class 3 and nine of class 2, all with centre and derived subgroup of order 4. For example, $C_2 \times D_{16}$ (SmallGroup(32, 39), class 3) and $(C_2 \times C_2) \wr C_2$ (SmallGroup(32, 27), class 2) both have commuting graph $K_4 \vee (K_{12} \sqcup 4 \cdot K_4)$, where $\sqcup$ denotes disjoint union. This is checked independently of the census in `tests/test_group_theory.py`. By Corollary 5.15 the two groups are not isoclinic. Element orders separate them, since $C_2 \times D_{16}$ has elements of order 8 and $(C_2 \times C_2) \wr C_2$ has exponent 4. In this dataset no R_1 class separates nilpotency class below order 64, and no R_2 class does so below order 128. Collisions of this kind follow from earlier results, although neither of the following papers discusses nilpotency class. Kakkar and Rawat [22, Cor. 2.2] show that the commuting graph of a generalized dihedral group $D(A)$ depends only on $|A|$ and the number of elements of order at most 2 in A. This gives the same graph for $D(C_8 \times C_2) \cong C_2 \times D_{16}$ and $D(C_4 \times C_4)$ (SmallGroup(32, 34), class 2). Pezzott and Nakaoka [20, Example 2.4] give a group J of order 32 with $J/Z(J) \cong C_2^3$, hence of class 2, whose commuting graph on a transversal of the centre is isomorphic to that of D_{16}. Since $C_2 \times D_{16}$ is isoclinic to D_{16} and has a centre of order $4 = |Z(J)|$, Lemma 5.10 turns this into a same-order collision between J and $C_2 \times D_{16}$.

At orders 32 and 64 the R_0 classes coincide with the classes of groups that have the same conjugacy class sizes (the same multiset of class sizes), which are tabulated on the Groupprops wiki [32]. The commuting graph determines the class sizes (Section 5), so the R_0 partition always refines this one, and at these two orders both partitions have the same number of classes (6 and 16). At order 128 the refinement is strict, with 46 R_0 classes against 44 class-size classes. For example, SmallGroup(128, 134) (class 5) and SmallGroup(128, 931) (class 4) have the same class sizes but non-isomorphic commuting graphs.

For the $p = 3$ dataset of 594 groups (orders 3 to 729), the census finds 33 R_0 classes covering 593 of the 594 groups (after refinement, 113 classes survive R_1 and 112 survive R_2). The pattern is qualitatively the same as for $p = 2$. The nilpotency-class floors are 0.1213, 0.0494 and 0.0340 at R_0, R_1 and R_2, the derived-length floor is 0 throughout, and the commutator-size floors are 0.0168, 0.0085 and 0.0049. At R_2, however, the nilpotency-class floor is much larger than at $p = 2$ (0.0340 against 0.0005). The powerfulness floor is 0.1397, 0.0606 and 0.0606, so power depth does not lower it at all (at $p = 2$ it falls from 0.0090 to 0.0045). The order-27 pair of Section 5.5 contributes only at R_0, since element orders separate it, so the floor at R_1 and R_2 comes from collisions of order at least 81.

Proposition 5.17 (Irreducible-error floor)**.** *Let $\mathcal{C}$ be a finite class of groups with the uniform distribution, R a representation, and I a real-valued invariant. Every predictor f that factors through R satisfies*

$$\mathbb{E}\big[(I(G) - f(R(G)))^2\big] \;\geq\; \mathbb{E}\big[\mathrm{Var}(I \mid R(G))\big],$$

with equality iff $f(R(G))$ is the mean of I over the R-class of G.

Proof. This follows from the conditional variance decomposition. For each R-class, the squared error of any constant prediction on that class is minimized by the class mean, with minimum value the class variance. Averaging over classes gives the result. □

By Proposition 5.17, the within-class variance of a target over the confirmed collision classes is a lower bound on the MSE of any graph-only model. The bound does not depend on the architecture

or on training, and it also applies with the persistent-homology branch, which, as noted above, is a function of the graph. Reporting this floor alongside Table 2 shows how much of each error is forced by the representation and how much is due to the model. Proposition 5.17 itself is the standard conditional-variance bound. For classification, Zopf [5] computes the analogous ceiling on accuracy for standard graph benchmarks, by a majority vote within each class of graphs that 1-WL does not distinguish. Here the classes are isomorphism classes rather than WL classes, so the floors apply to every model that reads the representation and not only to message-passing networks, and the targets are real-valued.

Remark 5.18 (Colour refinement and the closed-form targets). Colour refinement (1-WL) on the unlabelled commuting graph determines every vertex's degree after one round, hence the centralizer-size profile $\deg(x)+1$, hence $|Z(G)|$ (the number of universal vertices) and $k(G)$ (the average of the profile, Proposition 5.8). The two closed-form-recoverable targets are therefore 1-WL-computable, and so within the expressiveness class of GIN [3], even without the corresponding input features. Conversely, D_8 and Q_8 have isomorphic commuting graphs but different multisets of element orders, so the R_1 order labels strictly refine the graph-isomorphism type.

5.6 From Exact Recoverability to Statistical Predictability

Definition 5.19 (Statistical Predictability)**.** I is *statistically predictable* from R on $\mathcal{C}$ if there is a predictor f with $I(G) \approx f(R(G))$ with small error on $\mathcal{C}$.

The experiments in Section 7 compare invariants that are determined by a representation with invariants that are only statistically predictable and benefit from enrichment.

6 Training Procedure

Centre size and commutator size are stored and predicted as $\log_p$ of the raw size, which is an integer for p-groups, keeps the targets on a similar scale across orders and makes them comparable across primes for transfer. The closed-form baseline for centre size is computed in the same units. Each saved model stores its architecture (node, PH and output dimensions, and the pooling mode) together with the weights. Every evaluation rebuilds the model from this record, so a model trained with the PH branch is always evaluated with it. Targets are normalized by standard scaling fitted on the training set only, while node features are supplied to the network in their raw numerical units. We train with Adam (learning rate 10^{-3}) and Huber loss, an 80/10/10 train/val/test split stratified by group order, and early stopping with patience 20. Groups of order 128 make up 87% of the dataset, and stratification ensures that every order with at least 10 groups appears in the validation and test sets roughly in proportion to its size (orders 2, 4 and 8 are kept entirely in training). All random splits and initializations are seeded (`dataset.py`, `train.ipynb`).

7 Experimental Results

7.1 Prediction Accuracy

We trained the model five times (seeds 42–46) on the dataset of 2668 groups, on CPU only, with the persistent-homology branch enabled. This is the configuration used throughout (`use_ph=True` in `train.ipynb`). The split was 2134/266/268, stratified by order. Early stopping (patience 20) ended four runs after 37–55 epochs, and the fifth (seed 44) ran the full 200 epochs. Each run took 222–1132 s of wall-clock time (mean 451.5 s), with peak RSS at most 2.4 GB. The PH diagnostic

was the same in every run. The noncentral commuting graph has nontrivial H_1 for 490 of the 2624 non-abelian groups (18.4% of the dataset), and for the 44 abelian groups the graph is empty by convention. Centre and commutator size are reported in $\log_2$ units throughout, as stored. Table 2 reports pooled test MSE and rounded accuracy as mean $\pm$ sample std over the five seeds, and Table 3 breaks MSE down by group order. The per-seed results show that seed 42 is an outlier on every target except centre size, even though its best validation loss (0.213) was lower than that of seeds 45 and 46. Seed 42 is included in all averages. The variance across seeds is large, particularly for $k(G)$.

Table 2 also gives two baselines on the same splits, the training-set mean of each target and gradient boosting on per-graph summary statistics (Section 7.2). Averaged over the five seeds, the GNN has a higher MSE than the training-mean predictor on nilpotency class (1.73 against 0.53) and commutator size (1.05 against 0.72), a similar MSE on derived length, and a lower MSE on centre size and $k(G)$. These averages are dominated by seed 42, which is worse than the training-mean predictor on every target except centre size (Table 4). Over the other four seeds the mean GNN MSE is 0.29 on nilpotency class and 0.34 on commutator size, against 0.55 and 0.72 for the training mean on the same splits. Gradient boosting has a lower MSE than the GNN on every target at every seed. Several aspects of the training setup could contribute to this instability: the node features enter in raw units (element orders and centralizer sizes up to 128), the readout is mean pooling, only one hyperparameter configuration is used, and early stopping has patience 20. We did not run a controlled experiment to separate them, so we do not attribute the failure of seed 42, or the spread across the other seeds, to any one of them.

The stratified split is random at the group level. The census shows that in each of the five seeds all 268 test groups have an R_0-isomorphic counterpart in the training set, and 41–48% share their full R_2 representation class with a training group. Ivanov et al. [6] found the same effect, test graphs with isomorphic copies in the training set, across standard graph-classification benchmarks. Table 2 therefore measures interpolation to unseen groups whose representations were largely seen in training, and it does not measure generalization to unseen representations. The order-transfer experiment, which holds out entire orders, is the stronger test of generalization, and performance degrades there (Section 10). A split that is disjoint on representation classes is left for future work.

7.2 Ablation and Leakage Analysis

The ablations follow the hierarchy of Section 5, in which R_0 is the graph alone, R_1 adds element order and R_2 adds power depth. We also use a redundant-feature variant in which the R_0-determined labels (centralizer size, class size and centrality) are supplied as node features. The GNN rows of Table 4 use the following inputs:

- constant node features, so that the network sees only the graph (R_0),
- element order only (R_1),
- element order and power depth,
- element order and the R_0-redundant features,
- all five features (R_2),
- all five features together with persistent homology of the noncentral commuting graph (the main run).

Invariant	Closed form or R_2 floor	Train mean	Grad. boost.	GNN MSE	GNN acc.
Nilpotency class	floor 0.0005	0.53 ± 0.09	0.034 ± 0.005	1.73 ± 3.22	0.59 ± 0.10
Derived length	floor 0	0.022 ± 0.016	0.000	0.020 ± 0.019	0.99 ± 0.01
Centre ($\log_2$)	exact 0.000	0.90 ± 0.19	0.011 ± 0.004	0.45 ± 0.32	0.71 ± 0.16
Commutator ($\log_2$)	floor 0.0047	0.72 ± 0.12	0.014 ± 0.002	1.05 ± 1.61	0.73 ± 0.14
Conjugacy classes	exact 0.000	211.6 ± 70.4	0.000	188.5 ± 295.1	0.29 ± 0.30

Table 2: Test-set MSE on all nontrivial 2-groups of order at most 128 (2668 groups, stratified 80/10/10 split, 268 test groups), mean ± sample std over seeds 42–46. The second column is not a trained predictor. For centre size and $k(G)$ it gives the measured MSE of the zero-training closed-form identities (Propositions 5.7 and 5.8), which was 0.000 on the test set, up to floating-point precision, in every run. For the other three targets it gives the dataset-global census Bayes floor at R_2 (Proposition 5.17; for commutator size see also Corollary 5.16), a lower bound on the MSE over the whole dataset, not over a particular test split, for any model with R_2 input. The per-seed split-specific test floors vary with the split and can be higher or lower than these. "Train mean" predicts the training-set mean of each target, and gradient boosting uses per-graph summary statistics. The GNN columns are the `train.ipynb` configuration (R_2+PH), with rounded accuracy in the last column. The best single GNN run, seed 44, was selected post hoc on the test set and is shown for scale only. In row order it reached MSE 0.174, 0.012, 0.030, 0.029 and 0.246, with rounded accuracies 0.73, 0.98, 0.99, 0.97 and 0.82.

Order	Nilp. class	Der. length	Centre	Comm.	$k(G)$
16	0.56 ± 0.71	0.17 ± 0.24	0.59 ± 0.85	2.04 ± 1.61	138 ± 151
32	0.45 ± 0.32	0.10 ± 0.15	1.01 ± 0.67	1.73 ± 1.18	64 ± 70
64	0.42 ± 0.23	0.03 ± 0.04	0.66 ± 0.53	0.81 ± 0.63	39 ± 31
128	1.93 ± 3.71	0.02 ± 0.02	0.41 ± 0.34	1.05 ± 1.81	210 ± 336

Table 3: Test MSE by group order, mean ± std over the five seeds. The test set contains 2, 6, 27 and 233 groups of orders 16, 32, 64 and 128 respectively. Under the stratification rule, all groups of smaller order fall into the training set. Centre and commutator in $\log_2$ units.

The order-only and order-plus-redundant configurations carry the same information (level R_1), because the redundant labels are graph-derived. Their difference therefore measures the optimization benefit of supplying such labels as node features. Comparing the order-plus-redundant configuration with the all-five-feature configuration then isolates the incremental effect of power depth. Table 4 reports these ablations for a single seed (42) alongside the non-GNN baselines, all on the same stratified split. We also ran the baselines on all five seed splits, and the caption gives the five-seed mean and standard deviation for gradient boosting. Using the noncentral commuting graph itself as the input graph was outside the scope of this study.

Model (seed 42)	Nilp.	Der. len.	Centre	Comm.	$k(G)$
constant train-mean	0.458	0.019	0.923	0.720	230.7
linear regr. (summaries)	0.095	0.013	0.172	0.111	0.000
gradient boosting (summaries)	0.028	0.000	0.009	0.012	0.000
GNN, constant features (R_0)	0.155	0.000	0.017	0.032	0.481
GNN, element order only (R_1)	0.547	0.015	3.036	1.423	1866.2
GNN, order + power depth	1.131	0.016	1.723	0.856	668.3
GNN, order + R_0-redundant features	0.229	0.010	0.128	0.052	0.426
GNN, all five features (R_2)	0.248	0.016	0.299	0.185	1.103
GNN, R_2 + noncentral PH (= main run, seed 42)	7.492	0.053	0.521	3.917	705.5
dataset-global census floor at R_0	0.141	0	0	0.022	0

Table 4: Ablations and baselines, test MSE, single seed (42). The main-run row is the seed-42 run from Table 2, which is an outlier on every target except centre size, as noted in the discussion of that table. The comparison with and without PH is therefore available only at this outlier seed and is not a multi-seed estimate. Centre and commutator in $\log_2$ units. (i) At seed 42, gradient boosting on per-graph summary statistics beats every tested GNN configuration on all five targets, except that on derived length it ties the constant-feature GNN (0.000 each). Across all five seed splits it also beats the main R_2+PH run on every target. The five-seed gradient-boosting MSE (mean ± std) is 0.034±0.005, 0.000, 0.011±0.004, 0.014±0.002 and 0.000 in column order. Linear regression recovers $k(G)$ exactly, because $k(G)$ is the mean centralizer size by the Burnside identity, and mean centralizer size is one of its input features. (ii) On nilpotency class, the constant-feature GNN is within 13% of the applicable split-specific R_0 test floor (0.155 against 0.13765 on this seed's test set, whereas the dataset-global floor is 0.141). It also nearly recovers the two closed-form targets, as Remark 5.18 predicts (1-WL sees the centralizer profile through degrees alone). On representations seen in training it therefore comes close to the optimal class-mean prediction, which says nothing about generalization to unseen representations. (iii) No measured error is below the corresponding dataset-global or split-specific floor (seed-42 R_0 test floors: nilpotency 0.13765, commutator 0.02442, others 0). The constant-feature GNN is the configuration closest to a positive floor (item (ii)), and the main run is far above the floors on every target. (iv) The order+redundant row has lower MSE than the full-feature row on all five targets, by a wide margin on centre size, commutator size and $k(G)$. At this seed, supplying graph-derived redundant features as inputs gives a large optimization benefit (compare the order-only row), while adding power depth to them gives no measurable improvement, even though the census's R_1-to-R_2 floor refinement shows that power depth does carry information.

8 Powerful and Powerfully Nilpotent p-Groups

Section 2.2 recalled the definitions of powerful and powerfully nilpotent p-groups following Traustason and Williams [14, 17]. For every group in the dataset and for an arbitrary prime p, we compute whether the group is powerful and whether it is powerfully nilpotent, and if it is, its powerful nilpotence class and powerful coclass, by implementing the upper powerfully central series $\hat{Z}_n(G)$ (code in `generate_groups_sage.py`). We verified a pure-Python reimplementation of the same definitions (`tests/group_theory_lib.py`) against thirteen hand-constructed groups spanning $p = 2$ and $p = 3$ (order 4 to 27). It correctly classifies all abelian groups and the modular group M_{16} as powerful (with powerful class 1 and 2 respectively), and D_8, Q_8, D_{16}, SD_{16} and the order-27 extraspecial group of exponent 3 as not powerful.

8.1 A worked odd-prime example: powerful but not powerfully nilpotent

Extending the computation to $p = 3$ turns up a group worth working through by hand, since it is an instance of the divergence flagged in Section 2.2. Let $G = \mathbb{Z}_9 \rtimes \mathbb{Z}_3$ with presentation $\langle a, b \mid a^9 = b^3 = 1,\ bab^{-1} = a^4 \rangle$ (order 27, exponent 9).

Proposition 8.1. *G is powerful but not powerfully nilpotent.*

Proof. Powerful. Direct computation gives $[a, b] = a^{-1}(b^{-1}ab) = a^{-1}a^7 = a^6$ (since $4^{-1} \equiv 7 \pmod 9$), so $[G, G] = \langle a^6 \rangle = \langle a^3 \rangle$, of order 3. To see $G^3 = \langle a^3 \rangle$: the quotient $G/\langle a^3 \rangle$ is abelian of order 9 (as $[G, G] = \langle a^3 \rangle$) and is generated by two elements of order dividing 3, hence is $C_3 \times C_3$ of exponent 3. Therefore every cube g^3 lies in $\langle a^3 \rangle$, giving $G^3 \leq \langle a^3 \rangle$, and $a^3 \in G^3$ gives equality: $G^3 = \langle a^3 \rangle = [G, G]$. Hence $[G, G] \leq G^3$ and G is powerful (odd prime, so the defining inequality is against $G^p = G^3$, not G^4).

Not strongly powerful. $G^{p^2} = G^9 = \langle g^9 : g \in G \rangle$. Every element of G has order dividing 9 (the exponent of G), so $g^9 = 1$ for all g, giving $G^9 = \{1\}$. But $[G, G]$ has order $3 \neq 1$, so $[G, G] \not\leq G^9$: G is not strongly powerful.

Not powerfully nilpotent. G is 2-generator. By Proposition 2.1 ([17, Lemma 2.1]), every 2-generator powerfully nilpotent p-group is strongly powerful. Since G is not strongly powerful, it cannot be powerfully nilpotent. □

This agrees with our computation (in `tests/test_group_theory.py`), in which the upper powerfully central series $\hat{Z}_n(G)$ stalls at $\hat{Z}_1(G) = Z(G)$ (order 3) and never reaches all 27 elements. The computation is an exhaustive search over the 27-element multiplication table and the proof relies on Proposition 2.1, so the two are independent checks of the same fact. The fact itself is not new. Traustason and Williams [17, Sec. 4.1] remark, with a short proof, that no non-abelian p-group of order less than p^4 is powerfully nilpotent, which covers G. Together with the first part of the proof, this shows that their observation that every powerful 2-group is powerfully nilpotent [14] does not extend to odd primes. Here G serves as a proved test case for our computation of powerful class.

8.2 The powerful-group prediction experiment

Two observations motivate a prediction experiment on these invariants. First, powerful groups are rare, and their proportion falls as the order grows. Table 1 shows the powerfully nilpotent fraction falling from 60% at order 8 to 8.6% at order 64, and the direct count over the generated dataset extends the trend to 2.2% at order 128. Second, powerful nilpotence class and coclass are new, less-studied invariants. Unlike for centre size or $k(G)$, there is no known reason to expect a trivial closed-form shortcut, which makes them a more interesting test of the architecture. Proposition 8.1

also shows that the target is only partially defined, since the powerful class is undefined for groups that are powerful but not powerfully nilpotent, such as our worked example. A sound setup therefore needs a masked loss or a two-stage design (classify powerful, then regress class). (Whether the representation is separately more predictive on the rigid powerful subclass than on the non-powerful one is a further question, not evaluated here.)

In view of Proposition 8.1, the experiment used the two-stage design. A classification head for `is_powerful` was trained with binary cross-entropy and a positive-class weight fitted on the training split, and a regression head for `powerful_class` was trained under a mask, only on groups where the class is defined. The classification target is heavily imbalanced. Its base rate is $97/2668 = 3.6\%$ at $p = 2$, thinning to $51/2328$ at order 128, and $141/594 = 23.7\%$ at $p = 3$. We do not predict `powerful_coclass`, since coclass $= \log_p |G| -$ class (verified with zero violations on both datasets). Its predictions and errors are therefore derivable from those of the class head, and its MSE equals the class MSE by construction. At $p = 2$ every powerful group is powerfully nilpotent [14] and so has a defined powerful class. We checked this for every powerful group in the $p = 2$ dataset, and it held. At $p = 3$ the mask is necessary in practice. Of the 141 powerful 3-groups, 82 (58%) are not powerfully nilpotent, so groups like that of Proposition 8.1 make up most of the powerful 3-groups in the dataset. We trained the model with five seeds per prime, using the R_2 backbone and the same split protocol as before.

Table 5 gives the results. The weighted GNN classifier trades pooled accuracy for balanced accuracy, because it over-predicts the rare positive class. Gradient boosting on graph summaries is better on every classification metric and fits the masked regression closely (mean class MSE 0.050 at $p = 2$ and 0.000 at $p = 3$). Several aspects of the two-head GNN design could contribute to its weaker results: its trunk is shared with the weighted classification loss, the weighting of the masked loss depends on batch composition, and the node features enter in raw units, with values up to 729 at $p = 3$. As in Section 7, we did not test which, if any, of these is responsible. The census gives dataset-global lower bounds on the 0–1 misclassification rate for powerfulness of 0.0199/0.0090/0.0045 ($R_0/R_1/R_2$) at $p = 2$ and 0.1397/0.0606/0.0606 at $p = 3$. Gradient boosting reads summaries of the R_2 labels (element-order and power-depth counts), so the R_2 floors are the ones that bound it. Its five-seed mean test error rates, 0.013 at $p = 2$ and 0.100 at $p = 3$, are above these floors, and at $p = 2$ each seed's error is above its split-specific R_2 `is_powerful` test floor (0–0.004).

	Model	Acc.	Bal. acc.	F1	Class MSE (masked)
$p = 2$	majority class	0.964 ± 0.015	0.500	0.000	—
	GNN (two-head)	0.811 ± 0.144	0.827 ± 0.080	0.281 ± 0.108	med. 0.35 [0.28, 19.5]
	GBM (summaries)	0.987 ± 0.008	0.879 ± 0.069	0.786 ± 0.136	0.050 ± 0.069
$p = 3$	majority class	0.797 ± 0.046	0.500	0.000	—
	GNN (two-head)	0.760 ± 0.092	0.804 ± 0.096	0.599 ± 0.095	med. 1.71 [0.25, $6.7{\cdot}10^6$]
	GBM (summaries)	0.900 ± 0.024	0.844 ± 0.039	0.750 ± 0.058	0.000 ± 0.000

Table 5: Powerful-groups prediction, mean ± std over seeds 42–46 per prime (268 test groups at $p = 2$, 60 at $p = 3$). The classification metrics are the main result of this experiment, and the masked class regression is exploratory. Its test sets are tiny (6–16 defined groups per seed at $p = 2$, 1–6 at $p = 3$), and one $p = 2$ seed and two $p = 3$ seeds left clearly divergent regression heads. No seeds were excluded, and the GNN class-MSE entries are medians with ranges.

8.3 A further invariant: p-spectrum

A structurally different invariant that could serve as a further prediction target is an invariant of Caranti, Scoppola and Traustason [18], which they call the spectrum and we call the p-spectrum to make the prime explicit. For a finite p-group G of nilpotency class c, the i-th *layer* of the upper central series is $Z_i(G) \setminus Z_{i-1}(G)$, and the *$p$-spectrum* of G is

$$\{\, 1 \le i \le c : Z_i(G) \setminus Z_{i-1}(G) \text{ contains an element of order } p \,\}.$$

Unlike our five scalar targets, the p-spectrum is set-valued (equivalently, a length-c binary vector), making it a natural candidate for structured or multi-label prediction rather than regression. Caranti, Scoppola and Traustason characterize which subsets can occur:

Theorem 8.2 ([18, Thm. 1.4])**.** *Let p be a prime. If a finite p-group G has an element of order p in layer $k \ge 2$ of the upper central series, then its p-spectrum contains $\{1, \dots, \min(k, p-1)\}$. Conversely, for any $1 \le k \le p-1$ and $c \ge k$ there is a group of class c with p-spectrum exactly $\{1, \dots, k\}$, and for any $p \le c_1 < c_2 < \cdots < c_n \le c$ there is a group of class c with p-spectrum $\{1, \dots, p-1, c_1, \dots, c_n\}$.*

For $p = 2$, $\min(k, p-1) = 1$ for every $k \ge 1$, so Theorem 8.2 is least restrictive on the prime our main dataset uses. This is another reason why the odd-prime extension (Sections 8 and 10) is a more informative regime for this invariant than $p = 2$ alone.

The p-spectrum is defined via the ordinary upper central series, a different filtration from both the upper powerfully central series and the power series $G \ge G^p \ge G^{p^2} \ge \cdots$ that power depth tracks. The p-spectrum is not determined by the plain commuting graph: D_8 and Q_8 have isomorphic commuting graphs (Proposition 5.12) but 2-spectra $\{1, 2\}$ and $\{1\}$, since every noncentral element of Q_8 has order 4. Whether it is recoverable from R_1 or R_2, or requires information neither supplies, is an open question in the sense of our recoverability framework (Section 5).

9 Towards a Theoretical Framework

Our setup motivates several open questions.

9.1 Determination and Collision Problems

Which invariants are determined by $\Gamma(G)$, and how does adding filtration data (power depth) or commutator magnitude change what becomes recoverable? Defining $G \sim H$ iff $\Gamma(G) \cong \Gamma(H)$, what is the structure of the equivalence classes, and how do invariants vary within a class? Section 5.5 gives partial answers: isoclinism produces collisions systematically (Theorem 5.11), the exponent and powerfulness provably vary within collision classes (Corollary 5.13), and the five classical targets can only vary within a non-isoclinic collision class (Corollary 5.15). The commuting graph together with the order is not complete for the isoclinism family (the O'Brien order-64 and Arvind–Cameron order-128 collisions, Section 5.5). The remaining questions are which additional non-isoclinic collision classes occur, which of them survive the R_1/R_2 enrichments, and whether any of them separate nilpotency class or derived length. For groups of order at most 128, the census settles the last of these questions and determines which collision classes survive the R_1/R_2 enrichments. It does not test isoclinism, so it identifies a class as non-isoclinic only when one of the five targets varies within it (Corollary 5.15). Eick and Schanze address questions of the same flavour computationally, with a much richer invariant set, for the 10 494 213 groups of order 512 [31]. Proposition 5.17 converts the resulting collision classes into hard lower bounds on achievable loss for the affected examples.

9.2 Powerful-Group Perspective

Modern results on powerful and powerfully nilpotent groups [14, 15, 16, 17] suggest that filtrations such as the power series G^{p^k} and the upper powerfully central series encode structure not visible in $\Gamma(G)$ alone. This motivates statements of the form "without filtration data, invariant I cannot be recovered from commuting graphs," and conversely.

Williams [15] gives a reason to expect the power-depth feature (Section 3.2) to carry structural information.

Theorem 9.1 ([15, Thm. 12])**.** *Let G be a powerful p-group with p an odd prime, and let $N \leq G^p$ be normal in G. Then N is powerfully nilpotent.*

Corollary 9.2 ([15, Cor. 13])**.** *Let G be a powerful p-group and $i \in \mathbb{N}$. If N is normal in G^{p^i} and contained in $G^{p^{i+1}}$, then N is powerfully nilpotent ($G^{p^{i+2}}$ in place of $G^{p^{i+1}}$ when $p = 2$).*

By definition, $\mathrm{pdepth}(x)$ is the layer of the tower $G \geq G^p \geq G^{p^2} \geq \cdots$ that x survives to. Corollary 9.2 says that, for powerful G, any subgroup that is normal in G^{p^i} and contained in $G^{p^{i+1}}$ (in $G^{p^{i+2}}$ when $p = 2$) is powerfully nilpotent, whether or not G itself is (recall $\mathbb{Z}_9 \rtimes \mathbb{Z}_3$ from Section 8.1 is not, yet its own layers can still contain powerfully nilpotent pieces in this sense). So the power-depth feature, which we compute for every element, records the filtration to which Williams' results apply. This suggests a hypothesis, not tested in the present experiments, that node embeddings are more informative about local structure near power-depth boundaries than elsewhere. One could check it by probing the intermediate GIN layer activations, as well as the final pooled output, against power depth.

10 Broader Generalizations

The extensions below go beyond the main experiments. Most of them have not been attempted. The exception is the third, extension to odd primes and larger orders, which we have partly carried out and whose results are given in that item.

- **Non-commuting or commutator-weighted graphs.** The commutator-magnitude edge attribute $w(x, y) = \mathrm{ord}([x, y])$ is only informative on a graph that includes non-commuting pairs (Section 3.2). This needs (i) a modified generator that also records non-commuting pairs and their commutator order, and (ii) an edge-aware message-passing layer (e.g. GINE) in place of the edge-blind GIN convolution used here.

- **Lie rings via the Lazard correspondence.** Finite p-groups of nilpotency class less than p correspond to Lie rings via the Lazard correspondence, which Traustason and Williams use to construct powerfully nilpotent groups of maximal powerful class [16]. Learning on the corresponding Lie ring alongside the group's commuting graph would extend the powerful-group direction of Section 8. Since the correspondence needs class less than p, it applies only to abelian 2-groups, so this direction is aimed at odd primes.

- **Extension to odd primes and larger orders.** We ran two transfer experiments, one to a larger order and one to the prime 3. Order transfer (training the mean-pooled GIN on orders ≤ 64 and testing zero-shot on the 2328 order-128 groups) failed badly for the scale-dependent targets (centre 20.9 and commutator 11.0 in $\log_2$-MSE, $k(G)$ 2285), while the bounded targets had smaller absolute errors (nilpotency class 0.74, derived length 0.96), which are still large for targets that take only a few values. The $p = 3$ dataset ($1 + 2 + 5 + 15 + 67 + 504 = 594$

groups, orders 3 to 729) was generated by the same procedure with $p = 3$ in place of $p = 2$. Cross-prime fine-tuning (a 2-group R_2 checkpoint fine-tuned on 3-groups at learning rate 10^{-4}) reached test MSE 0.259/0.001/0.214/0.227/28.1 on the five targets. We compared this with training from scratch at the standard learning rate 10^{-3} (same seed, split, and epoch budget), which reached 0.466/0.000/0.534/0.426/2390.9. Fine-tuning was ahead on four of the five targets, by the widest margin on $k(G)$ (28 vs. 2391), and training from scratch was marginally ahead only on derived length. This is a single-seed comparison and does not establish a general fine-tuning advantage. A from-scratch run at the fine-tuning learning rate 10^{-4} diverged, with MSEs of order 10^4–10^{11}. The SmallGroups library contains all groups of order at most 2000 except 1024, all groups of order p^n for $n \leq 6$, and all groups of order p^7 for $p = 3, 5, 7, 11$ [2], so larger odd-prime datasets (for example order 3^7) can be generated in the same way.

- **Coclass trees.** Placing sampled groups inside coclass trees as an explicit structural feature was not investigated in this study.

11 Conclusion

We built a dataset and GNN pipeline for predicting classical invariants of finite p-groups from enriched commuting graphs, and a formal framework (Section 5) for asking which invariants a given graph representation can determine exactly. Two of the five targets we predict, centre size and the number of conjugacy classes, turn out to be exact, closed-form functions of the node features already supplied to the model (Propositions 5.7 and 5.8). We report the closed-form values as a baseline, since they account for any strong performance on these two targets.

Section 5.5 adds negative results to the recoverability framework. They rest on two known results: same-order isoclinic groups share a commuting graph (Pezzott and Nakaoka [20]), and the two non-abelian groups of order p^3 form an infinite collision family [21, 24]. From these we derive that the exponent and powerfulness are not recoverable from the unlabelled commuting graph (Corollary 5.13), that $\tau(\exp) = 1$, and that damaging collisions are necessarily non-isoclinic (Corollary 5.15). Using the order-128 example of Arvind and Cameron [25], we prove that commutator size, one of the five classical targets, has a strictly positive irreducible error floor for unlabelled-graph (R_0) models on this dataset (Corollary 5.16, Proposition 5.17). The census confirms that this floor survives the R_1 and R_2 enrichments used in our experiments (0.0047 in $\log_2$ units at R_2, Section 7). It also settles empirically, up to order 128, the two recoverability cases left open in Section 5.5. Nilpotency class is separated by non-isoclinic collisions, with a Bayes floor of 0.0005 even at R_2, while derived length is separated by none. The smallest such collision has order 32, where $C_2 \times D_{16}$ and $(C_2 \times C_2) \wr C_2$ share a commuting graph. Element orders separate this pair, and the R_2 floor comes from collisions of order 128. The class-2 conjecture of Arvind and Cameron [25] was refuted by O'Brien's order-64 examples [24], and the census determines which of our targets such collisions damage.

Section 8 uses the framework of Traustason and Williams [14, 17] to extend the experiments to a new, less-studied target, the powerful nilpotence class, which we compute for an arbitrary prime p. Generalizing to odd primes produced an example (Proposition 8.1) of a powerful group that is not powerfully nilpotent, a special case of a remark of Traustason and Williams [17, Sec. 4.1], and this example gives an independent check on the odd-prime computations. Corollary 9.2 shows that the power-depth feature records the filtration to which Williams' results apply. This is a structural connection, not evidence that the feature improves prediction. The evidence that power depth carries information about our targets comes from the census, where the nilpotency-class floor falls from 0.0090 at R_1 to 0.0005 at R_2 and the commutator-size floor from 0.0066 to 0.0047.

In our single-seed ablation, adding power depth to the order-plus-redundant GNN input gave no improvement, and adding it to element order alone gave mixed results (Table 4). The p-spectrum of Caranti, Scoppola and Traustason [18] (Section 8.3) is proposed as a further, structurally different target for future work. The plain commuting graph does not determine it, and its recoverability from R_1 and R_2 is left open.

The exhaustive census recovered both validation cases from the literature. Three predictions of the theory are borne out in the experiments. The closed-form identities scored zero error, up to floating-point precision, in every run. The graph-only ablation landed within 13% of the split-specific R_0 test floor on nilpotency class (0.155 vs. 0.13765 on its seed's test set), which is near-optimal class-mean recovery within seen representations. As the 1-WL analysis of Remark 5.18 anticipates, the same ablation also nearly recovered the two closed-form targets. No measured error is below its floor, but most are far above their floors. In particular, the main R_2+PH model, averaged over five seeds, is worse than predicting the training mean on nilpotency class and commutator size, mainly because of one failed seed (Section 7).

The runs also give a negative result about our GNN pipeline. Gradient boosting on per-graph summary statistics outperformed the GNN on every target across all five seed splits, and per-seed variance under early stopping was large. Because the node features enter in raw units and only one hyperparameter configuration was tried, this does not show that graph neural networks on commuting graphs fail in general. The split is random at the level of groups, so the pooled numbers measure interpolation within largely seen representations (every test group has an R_0-isomorphic training partner), and zero-shot extrapolation to unseen larger orders failed for every scale-dependent target. The Bayes floors bound the error of any model that sees only these representations, and the gap between the GNN's errors and these floors measures what this model does not achieve. Cross-prime fine-tuning was the transfer setting in which the architecture looked strongest: in a single-seed comparison it beat training from scratch at the standard learning rate on four of the five targets (Section 10).

The powerful-groups extension was also evaluated (Table 5). Proposition 8.1 makes a masked two-stage design necessary, and on the 3-group data the mask was essential, since 58% of powerful 3-groups are not powerfully nilpotent. On the powerfulness classification the enriched-graph GNN beats the majority-class baseline on balanced accuracy and F1, though not on raw accuracy, and it does not beat gradient boosting on summary statistics. The masked class-regression estimates are exploratory because the eligible test subsets are small (see the caption of Table 5).

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Use of AI tools

AI tools were used for programming and for checking computations.

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